

Midterm 2

Grant Talbert

November 22, 2025

Boston University

Part I Quickies

1 Plotting momentum space wavefunction: harmonic oscillator

The first excited momentum space wavefunction $\phi_1(p) = \langle p | 1 \rangle$ is

$$\phi_1(p) = \left(\frac{1}{\pi m \hbar \omega} \right)^{1/4} \frac{p \sqrt{2}}{\sqrt{m \hbar \omega}} \exp \left(-\frac{p^2}{2m \hbar \omega} \right).$$

Let $p_0 = \sqrt{m \omega \hbar}$ to simplify notation, such that

$$\phi_1(p) = \left(\frac{1}{\pi p_0^2} \right)^{1/4} \frac{p \sqrt{2}}{p_0} \exp \left(-\frac{p^2}{2p_0^2} \right).$$

Then

$$|\phi_1(p)|^2 = \frac{1}{\sqrt{\pi p_0^2}} \frac{2p^2}{p_0^2} \exp \left(-\frac{p^2}{p_0^2} \right) = \frac{2p^2}{p_0^3 \sqrt{\pi}} \exp \left(-\frac{p^2}{p_0^2} \right).$$

We set $m = \omega = \hbar = 1$ in our plot.

```
1 #!/bin/python3
2
3 import numpy as np
4 import matplotlib.pyplot as plt
5
6 # natural units
7 m = 1.0
8 omega = 1.0
9 hbar = 1.0
10
11 p0 = np.sqrt(m * hbar * omega)
12
13 p = np.linspace(-5 * p0, 5 * p0, 1000)
14
15 prob_density = 2 * (p**2) * np.exp(-(p**2) / (p0**2)) / ((p0**3) * np.sqrt(np.pi))
16
17 # plotting
18 plt.figure(figsize=(8, 6))
19 plt.plot(p, prob_density, color='black', linewidth=1.5)
20
21 plt.xlabel(r'$p = \hbar k$', fontsize=14)
22 plt.ylabel(r'$|\langle p | 1 \rangle|^2$', fontsize=14)
23 plt.title(r'Momentum Space Probability Density: First Excited State ($n=1$)',
24           fontsize=14)
25
26 plt.grid(True, linestyle='--', alpha=0.7)
27 plt.tight_layout()
28 plt.savefig('asnjddsnfjk.png')
```

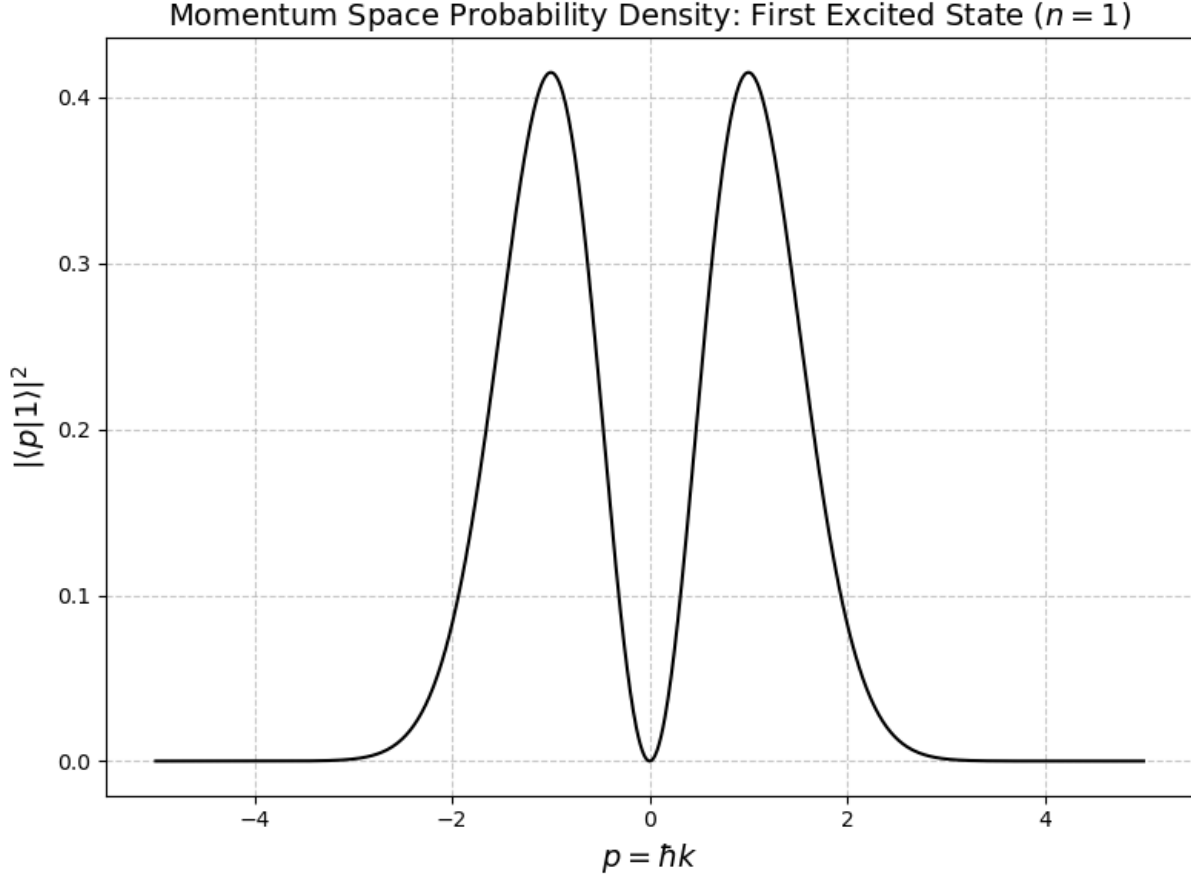


Figure 1: Momentum space probability density for first excited state $|n\rangle = |1\rangle$

2 Interaction potential

We use Hadamard's operator lemma

$$e^A B e^{-A} = B + [A, B] + \frac{1}{2}[A, [A, B]] + \dots$$

with $A = iH_0 t/\hbar$ and $B = V$. We then use the identities

$$[AB, C] = A[B, C] + [A, C]B,$$

$$[A, BC] = [A, B]C + B[A, C]$$

to compute

$$[\hat{x}^2, \hat{p}^2] = \hat{x}[\hat{x}, \hat{p}^2] + [\hat{x}, \hat{p}^2]\hat{x} = \hat{x}([\hat{x}, \hat{p}]\hat{p} + \hat{p}[\hat{x}, \hat{p}]) + ([\hat{x}, \hat{p}]\hat{p} + \hat{p}[\hat{x}, \hat{p}])\hat{x} = 2i\hbar\hat{x}\hat{p} + 2i\hbar\hat{p}\hat{x}.$$

We then have

$$[iH_0 t/\hbar, V] = \left[\frac{it\hat{p}^2}{2m\hbar}, \frac{m\omega_0^2 \hat{x}^2}{2} \right] = \frac{it\omega_0^2}{4\hbar} [\hat{p}^2, \hat{x}^2] = -\frac{it\omega_0^2}{4\hbar} [\hat{x}^2, \hat{p}^2] = \frac{t\omega_0^2}{2} \{\hat{x}, \hat{p}\}$$

for $\{-, -\}$ the anticommutator. We then compute

$$[iH_0t/\hbar, [iH_0t/\hbar, V]] = \frac{t\omega_0^2}{2} \left[\frac{it\hat{p}^2}{2m\hbar}, \{\hat{x}, \hat{p}\} \right] = \frac{it^2\omega_0^2}{4m\hbar} [\hat{p}^2, \{\hat{x}, \hat{p}\}].$$

We compute the commutator independently in order to not keep rewriting the coefficient.

$$[\hat{p}^2, \hat{p}\hat{x} + \hat{x}\hat{p}] = [\hat{p}^2, \hat{p}\hat{x}] + [\hat{p}^2, \hat{x}\hat{p}] = [\hat{p}^2, \hat{p}]\hat{x} + \hat{p}[\hat{p}^2, \hat{x}] + [\hat{p}^2, \hat{x}]\hat{p} + \hat{x}[\hat{p}^2, \hat{p}] = \hat{p}[\hat{p}^2, \hat{x}] + [\hat{p}^2, \hat{x}]\hat{p}.$$

We can compute $[\hat{p}^2, \hat{x}]$:

$$[\hat{p}^2, \hat{x}] = \hat{p}[\hat{p}, \hat{x}] + [\hat{p}, \hat{x}]\hat{p} = -2i\hbar\hat{p}.$$

We thus obtain

$$[\hat{p}^2, \{\hat{p}, \hat{x}\}] = \hat{p}(-2i\hbar\hat{p}) + (-2i\hbar\hat{p})\hat{p} = -4i\hbar\hat{p}^2.$$

The full term is thus

$$\frac{1}{2} \frac{it^2\omega_0^2}{4m\hbar} (-4i\hbar)\hat{p}^2 = \frac{t^2\omega_0^2}{2m}\hat{p}^2.$$

Since $\hat{p}$ commutes with itself, all higher order terms vanish, so we obtain

$$V_I = \frac{m\omega_0^2}{2}\hat{x}^2 + \frac{t\omega_0^2}{2}\{\hat{x}, \hat{p}\} + \frac{t^2\omega_0^2}{2m}\hat{p}^2.$$

3 Josephson-junction

We are given $V = v_0 \cos \omega t$. Applying this to the time-evolution of $\phi_2 - \phi_1$ gives

$$\frac{\partial(\phi_2 - \phi_1)}{\partial t} = \frac{2e}{\hbar} v_0 \cos \omega t.$$

Integrating and assuming the junction has no initial phase difference yields

$$\phi_2 - \phi_1 = \frac{2e}{\hbar} \frac{\hbar\omega}{2e} \int \cos \omega t \, dt = \sin \omega t.$$

We then obtain

$$I(t) = I_c \sin(\sin(\omega t)).$$

We plot the current $I(t)$ as a function of $\tau = \omega t$ over two periods. We take $I_c = 1$ for simplicity.

```

1 #!/bin/python3
2
3 import numpy as np
4 import matplotlib.pyplot as plt
5
6 # take tau to be omega*t
7 tau = np.linspace(0, 4 * np.pi, 1000)
8
9 # take I_c=1
10 current = np.sin(np.sin(tau))
11
12 # 4. Plotting
13 fig, ax = plt.subplots(figsize=(8, 6))
14
15 ax.plot(tau, current, 'k-', linewidth=1.5, label=r'$I(t) = I_c \sin(\sin(\omega t))$')
16

```

```

17 ax.set_xlabel(r'$\omega t$ (rad)', fontsize=12)
18 ax.set_ylabel(r'Current $I(t)$', fontsize=12)
19
20 # just make it look a lil nicer
21 ax.axhline(0, color='gray', linestyle='--', linewidth=0.8, alpha=0.7)
22 ax.grid(True, which='both', linestyle=':', alpha=0.6)
23 ax.set_xlim(0, 4 * np.pi)
24 ax.set_ylim(-1.1, 1.1)
25 ax.legend(loc='upper right', frameon=True)
26
27 plt.savefig('josephson_current.png', dpi=300)
28 plt.show()

```

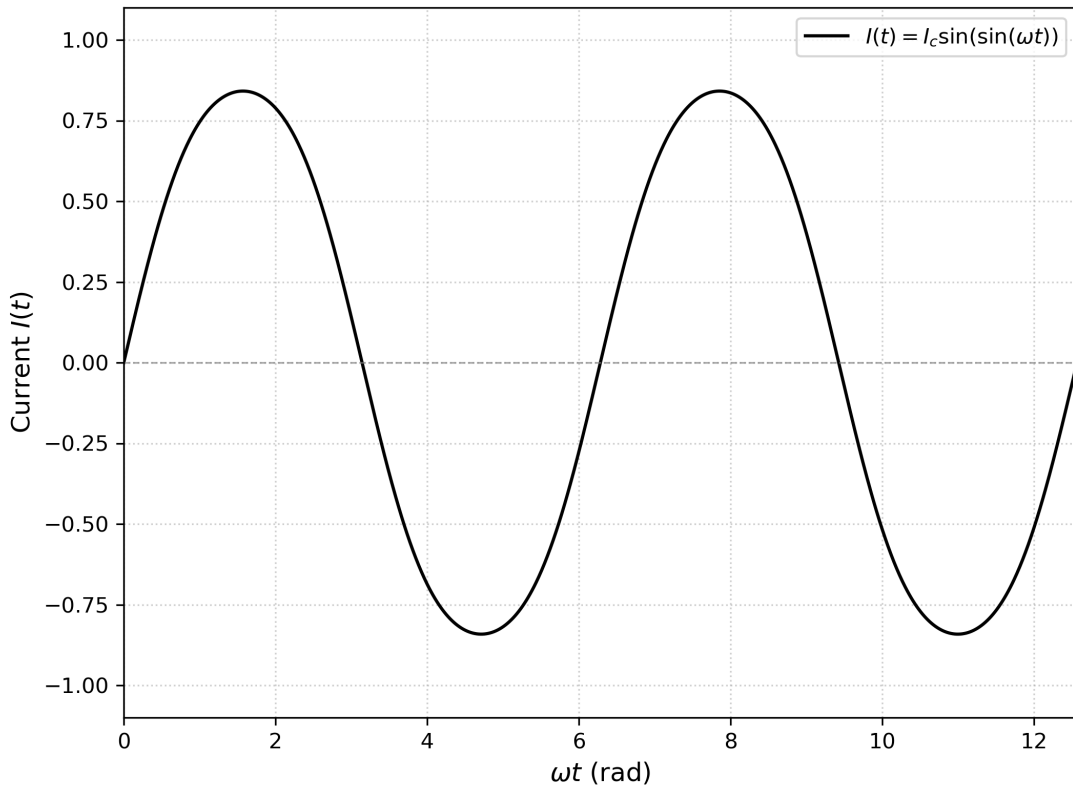


Figure 2: The current I of the Josephson junction plotted along 2 periods against ωt .

The function $\sin(\sin(\omega t))$ is an odd periodic function, as seen in the figure, and its average value over one period is also seen to be 0. Thus there is no DC current.

4 Displacement operator

(i) Alfie's translation operators send $\hat{D}(-\alpha)|\alpha\rangle = |\alpha - \alpha\rangle = |0\rangle$ and $\hat{D}(-\beta)|0\rangle = |-\beta\rangle \neq |\beta\rangle$. Therefore Alfie's setup is wrong.

(ii) Betty deserves full credit. Since $\hat{D}$ is unitary, its inverse is given by $\hat{D}^\dagger$. Since $\hat{D}(\alpha)|0\rangle = |\alpha\rangle$,

its inverse acts by $\hat{D}^\dagger|\alpha\rangle = |0\rangle$. We then have $\hat{D}(\beta)|0\rangle = |\beta\rangle$. Therefore, Betty's transformations indeed map $|\alpha\rangle \mapsto |\beta\rangle$.

(iii) Gammy's setup is also wrong.

$$\hat{D}(\alpha)\hat{D}(\beta)|\alpha\rangle = \hat{D}(\alpha)|\beta + \alpha\rangle = |2\alpha + \beta\rangle \neq |\beta\rangle.$$

(iv) Delter's setup is correct. Noting again that $\hat{D}^\dagger(z) = \hat{D}^{-1}(z) = \hat{D}(-z)$ for any z ,

$$\hat{D}^\dagger(-\beta)\hat{D}^\dagger(\alpha)|\alpha\rangle = \hat{D}(\beta)\hat{D}(-\alpha)|\alpha\rangle = \hat{D}(\beta)|0\rangle = |\beta\rangle.$$

(v) Epsom's setup is wrong.

$$\hat{D}^\dagger(\alpha)\hat{D}^\dagger(\beta)|\alpha\rangle = \hat{D}(-\alpha)\hat{D}(-\beta)|\alpha\rangle = \hat{D}(-\alpha)|\alpha - \beta\rangle = |-\beta\rangle \neq |\beta\rangle.$$

5 Quantum LC circuit

(i) In an LC circuit the angular frequency is $\omega = \frac{1}{\sqrt{LC}}$. This yields $C = \frac{1}{\omega^2 L}$. We have $\omega = 2\pi f = 2\pi(5 \cdot 10^9) = 10\pi \cdot 10^9 \text{ rad/s}$. Then

$$C = \frac{1}{(10^{-8}\text{H})(10\pi \cdot 10^9 \text{ rad/s})^2} = \frac{1}{10^{12}\pi^2} \text{F} \approx 1 \cdot 10^{-13} \text{F}.$$

(ii) No.

(iii) No, due to the uncertainty principle. There is a conjugation relation $[\hat{\Phi}, \hat{Q}] = i\hbar$, and thus an uncertainty principle

$$\Delta\Phi\Delta Q \geq \frac{\hbar}{2}$$

is satisfied. To detect a cooper pair, the state would need to be in a Fock state of charge with small ΔQ . To detect a flux quanta, the state would need to be in a Fock state of flux with small $\Delta\Phi$. The uncertainty principle forbids both quantities being small.

Part II

Longer problems

6 Simple harmonic oscillator with perturbation in momentum

(a) Because $\hat{H}$ is the *energy* operator, $\lambda\hat{p}$ must have units of energy, meaning λ must have units of velocity. For the perturbation to be *weak*, we would want the induced energy shift to be significantly smaller than the energy gap between states. In the 1D harmonic oscillator with a nonperturbed Hamiltonian, the energy difference is $\Delta E = \hbar\omega_0$, so we want the perturbation to induce a much smaller energy change than this. Recall this is approximately $\lambda\sqrt{m\hbar\omega_0}$, so we have

$$\lambda \ll \frac{\hbar\omega_0}{\sqrt{m\hbar\omega_0}} = \sqrt{\frac{\hbar\omega_0}{m}}.$$

(b) Start by rewriting the momentum-dependent terms of the Hamiltonian as

$$\frac{\hat{p}^2}{2m} + \lambda\hat{p} = \frac{1}{2m} (\hat{p}^2 + 2m\lambda\hat{p}).$$

We can complete the square:

$$= \frac{1}{2m} (\hat{p}^2 + 2m\lambda\hat{p} + m^2\lambda^2 - m^2\lambda^2) = \frac{1}{2m} (\hat{p} + m\lambda)^2 - \frac{m\lambda^2}{2}.$$

Substituting this back into the Hamiltonian, we have

$$\hat{H} = \frac{1}{2m} (\hat{p} + m\lambda)^2 - \frac{m\lambda^2}{2} + \frac{1}{2}m\omega_0^2\hat{x}^2.$$

We define

$$\begin{aligned}\hat{X} &:= \omega_0\sqrt{m}\hat{x} \\ \hat{P} &:= \frac{\hat{p} + m\lambda}{\sqrt{m}} \\ \text{constant} &:= -\frac{m\lambda^2}{2}.\end{aligned}$$

The Hamiltonian then takes the form

$$\hat{H} = \frac{1}{2}\hat{P}^2 + \frac{1}{2}\hat{X}^2 + \text{constant}.$$

(c)

$$\begin{aligned}[\hat{X}, \hat{P}] &= \left[\omega_0\sqrt{m}\hat{x}, \frac{\hat{p}}{\sqrt{m}} + \frac{m\lambda}{\sqrt{m}} \right] \\ &= \left[\omega_0\sqrt{m}\hat{x}, \frac{\hat{p}}{\sqrt{m}} \right] + \left[\omega_0\sqrt{m}\hat{x}, \frac{m\lambda}{\sqrt{m}} \right] \\ &= \left[\omega_0\sqrt{m}\hat{x}, \frac{\hat{p}}{\sqrt{m}} \right] \\ &= \omega_0 [\hat{x}, \hat{p}] \\ &= i\hbar\omega_0.\end{aligned}$$

(d) Because the operators $\hat{X}, \hat{P}$ have the commutation relation given in part (c), they must describe a harmonic oscillator with the same frequency ω_0 as the unperturbed system. The energy eigenvalues are therefore the exact same as the standard oscillator eigenvalues

$$E_n = \hbar\omega_0 \left(n + \frac{1}{2} \right).$$

The total energy of the perturbed system is thus just shifted by the constant term

$$E_n = \hbar\omega_0 \left(n + \frac{1}{2} \right) - \frac{m\lambda^2}{2}.$$

The correction in any state induced by this perturbation is thus

$$\Delta E_n = -\frac{m\lambda^2}{2}.$$

This includes the $n = 1$ case.

7 Landau problem, symmetric gauge

(a)

$$\mathbf{B} = \nabla \times \mathbf{A} = \left(\partial_y 0 - \partial_z \frac{1}{2} B_0 x \right) \hat{\mathbf{x}} - \left(\partial_x 0 + \partial_z \frac{1}{2} B_0 y \right) \hat{\mathbf{y}} + \left(\partial_x \frac{1}{2} B_0 x + \partial_y \frac{1}{2} B_0 y \right) \hat{\mathbf{z}} = B_0 \hat{\mathbf{z}}$$

(b) The Hamiltonian in the xy -plane will be given by

$$\begin{aligned} \hat{H} &= \frac{1}{2m} (\hat{\mathbf{p}} - q\mathbf{A})^2 \\ &= \frac{1}{2m} \left(\left(\hat{p}_x + \frac{qB_0 y}{2} \right)^2 + \left(\hat{p}_y - \frac{qB_0 x}{2} \right)^2 \right) \\ &= \frac{1}{2m} \left(\hat{p}_x^2 + \frac{q^2 B_0^2 y^2}{4} + qB_0 y \hat{p}_x \right) + \left(\hat{p}_y^2 + \frac{q^2 B_0^2 x^2}{4} - qB_0 x \hat{p}_y \right) \\ &= \frac{\hat{p}_x^2 + \hat{p}_y^2}{2m} + \frac{q^2 B_0^2}{8m} (x^2 + y^2) + \frac{qB_0}{2m} (y\hat{p}_x - x\hat{p}_y). \end{aligned}$$

Note the angular momentum operator $\hat{L}_z = x\hat{p}_y - y\hat{p}_x$, and define $\omega := \frac{qB_0}{2m}$. The Hamiltonian then reads as

$$\hat{H} = \frac{\hat{\mathbf{p}}^2}{2m} + \frac{1}{2} m \omega^2 (x^2 + y^2) - \omega \hat{L}_z.$$

The first two terms resemble that of a 2D oscillator with frequency ω , but there is an extra $\omega \hat{L}_z$ term at the end.

(c) The Hamiltonian becomes

$$\hat{H} = \frac{\hat{\mathbf{p}}^2}{2m} + \frac{1}{2} m \omega^2 r^2 - \omega \hat{L}_z.$$

Recall the Laplacian in polar coordinates in 2D is

$$\nabla^2 = \frac{\partial^2}{\partial r^2} + \frac{1}{r} \frac{\partial}{\partial r} + \frac{1}{r^2} \frac{\partial^2}{\partial \phi^2}.$$

Additionally,

$$\hat{L}_z = -i\hbar \frac{\partial}{\partial \phi}.$$

The time-dependent Schrödinger equation is thus

$$i\hbar \frac{\partial}{\partial t} \Psi(x, y, t) = \left(-\frac{\hbar^2}{2m} \left(\frac{\partial^2}{\partial r^2} + \frac{1}{r} \frac{\partial}{\partial r} + \frac{1}{r^2} \frac{\partial^2}{\partial \phi^2} \right) + \frac{1}{2} m \omega r^2 + i\hbar \omega \frac{\partial}{\partial \phi} \right) \Psi(x, y, t).$$

The time-independent Schrödinger equation is

$$\left(-\frac{\hbar^2}{2m} \left(\frac{\partial^2}{\partial r^2} + \frac{1}{r} \frac{\partial}{\partial r} + \frac{1}{r^2} \frac{\partial^2}{\partial \phi^2} \right) + \frac{1}{2} m \omega r^2 + i\hbar \omega \frac{\partial}{\partial \phi} \right) \psi(x, y) = E \psi(x, y).$$

(d) There is a degeneracy. The usual oscillator has energy eigenvalues $\hbar\omega(2n + |m| + 1)$, but since $\hat{L}_z$ has an eigenvalue of $m\hbar$, the $-\omega \hat{L}_z$ term shifts the eigenvalues to

$$\hbar\omega(2n + |m| + 1) - \omega m \hbar.$$

Taking $n = 0$ and $m \geq 0$ yields $E_0 = \hbar\omega$. We let m range over all integers, and in particular positive integers, and we have shown for each positive integer there is a ground state with the same energy eigenvalue $\hbar\omega$.